

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Vector Operations: Dot Product & Cross Product	36 PYQs	Vector Operations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Vector Operations: Dot Product & Cross Product	$\vec{a} \cdot \vec{b} = 0$ iff $\vec{a} \perp \vec{b}$, $\text{Projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} }$, $\text{Area} = \frac{1}{2} \vec{a} \times \vec{b} $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Vector Operations: Dot Product & Cross Product (Very High Yield — 36 PYQs)

Core Principle: Dot product $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$. Two non-zero vectors are perpendicular if and only if $\vec{a} \cdot \vec{b} = 0$. Projection of $\vec{a}$ on $\vec{b}$: $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$. Cross product $\vec{a} \times \vec{b}$ gives a vector perpendicular to both $\vec{a}$ and $\vec{b}$; two vectors are parallel if $\vec{a} \times \vec{b} = \vec{0}$. Area of triangle = $(1/2)|\vec{a} \times \vec{b}|$; area of parallelogram = $|\vec{a} \times \vec{b}|$.

■ **Caution / JEE Pitfall:** Projection of $\vec{a}$ on $\vec{b}$ has $|\vec{b}|$ in denominator, NOT $|\vec{a}|$!

■ **Speed Trick:** Lagrange's Identity: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$. Useful for instant evaluation in KCET.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a}=2\hat{i}+\lambda\hat{j}+k\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a}=i\hat{i}+j\hat{j}+k\hat{k}$, $\vec{b}=j\hat{i}-k\hat{k}$ and $\vec{a} \times \vec{c}=\vec{b}$, $\vec{a} \cdot \vec{c}=3$, then $\vec{c}$ i..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $\vec{a}=2i\hat{i}+2j\hat{j}-k\hat{k}$, $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2k\hat{k}$ and $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, then $\alpha+\beta=...$ " [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value of λ for which the vectors $\text{vec}(\mathbf{a}) = 2\mathbf{i} + \lambda\mathbf{j} + \mathbf{k}$ and $\text{vec}(\mathbf{b}) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ are orthogonal is

- (A) $(5 / 2)$
- (B) $(-5 / 2)$ **[Correct]**
- (C) $(2 / 5)$
- (D) $(-2 / 5)$

Correct Answer: B

Solution Analysis: For two vectors to be orthogonal, their dot product must be zero. Given, $\text{vec}(\mathbf{a}) = 2\mathbf{i} + \lambda\mathbf{j} + \mathbf{k}$ and $\text{vec}(\mathbf{b}) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Now, $\text{vec}(\mathbf{a}) \cdot \text{vec}(\mathbf{b}) = 2(1) + \lambda(2) + 1(3) = 2 + 2\lambda + 3 = 5 + 2\lambda$. Since the vectors are orthogonal, $5 + 2\lambda = 0$
 $2\lambda = -5$ $\lambda = (-5 / 2)$ So, the correct option is Option B $(-5 / 2)$